DRBD

Guide	DRBD, HA+DR, primary and secondary, drbdproxy, cluster_install.sh, coro_install, silo-cluster-install, CRM templates, crm-template-updater.sh, logical block sizes do not match
Team Owner	SysEng
Status	OUTDATED
Last Modified	Apr 28, 2020

- Introduction
- Diagnosis and Mitigation
  - drbdadm status
  - DB / Collector connectivity check
  - systemctl status em7
  - licensed state check
  - ha status check
- Theory
  - file:/var/lib/em7/stats/sl1/licensed_state
  - file:/var/lib/em7/stats/drbd_data/drbd_cs
  - file:/opt/em7/share/config/coro_install.d/drbd_v9/drbd_proxy
  - file:/opt/em7/share/config/coro_install.d/cib_stacked
  - CRM Templates
  - program:crm-template-updater.sh
- Post Mortem Articles
  - 2025-06-13 EM-74609 RFO 00519148 - ongoing cpu issues after outage related to DR replication and DRBD network failures
  - 2024-10-30 EM-70932: DRBD constantly sync restarts at 0% after upgrade OL7 to OL8
  - 2024-08-26 EM-66540: block drbd1: logical block size of local backend does not match (drbd:512, backend:4096)
  - 2024-08-22 EM-66512: DR replication to DR node stops after HA failover
  - 2024-08-20 EM-66492: em7 service gets hung after HA failover, doesn't restart for 15 minutes delaying other necessary services from starting
  - 2024-08-13 EM-66389: Sigterms and UI doesn't fully work after failover
  - 2024-06-04 EM-65272: Setting up HA/DRBD

- 2024-05-09 EM-64907: CRM Template Outdated blocker for OL8 conversion
- 2024-05-08 EM-64840 Issue after update CRM template
- 2024-04-17 EM-64479: silo-cluster-install failes to create DRBD metadata on secondary and DR nodes,
- 2024-04-15 EM-64415 /root/.cluster_install/r0.res:126: Parse error: 'ssocks | sdp | ipv4 | ipv6 | <ipv4 address> ' expected, but got ':'
- 2024-04-15 EM-64413: SNMP reporting that one DB is not licensed and another is not in sync, neither of those appear to be true
- 2024-04-15 EM-64409: coro_install.d/drbd_v9/drbd_proxy template is incorrect
- 2024-04-04 EM-64174: Silo-cluster-install script fails to connect to the remote database when the password contain special chars
- 2024-04-04 EM-64157: DRBD replication does not work after phase 2 due to "The peer's disk size is too small!"
- 2024-03-27 EM-64054: DRBD is triggered and becomes out-of-sync. This process gets hung and doesn't complete.
- 2024-02-09 EM-63276 Customer Needs Advice on Resizing DB disk space beyond 2TB with special circumstances
- 2024-01-31 EM-63149: Coro_config not working after 12.1.1. update
- 2020-04-28 Symantech

Introduction

This page has more detailed information on DRBD debugging: Troubleshooting DRBD OL7/DRBD8 . The post mortem articles section of this guide, however, will still be used to gather commentary on escalations related to drbd as they occur.

Diagnosis and Mitigation

drbdadm status

```
drbdadm status
```

DB / Collector connectivity check

You can run

```
/opt/em7/backend/collector_check.py
```

to verify that the active DB can reach all collectors.

systemctl status em7

This can discover problems with the clustering setup

```
 systemctl status em7
```

Here is an example output from a system that had DRBD issues:

```
System Status v3.72
Wed May  1 20:45:02 UTC 2024

[1;48;5;28mResults marked with [KB ####] refer to a KB article number you can reference at https://support.sciencelogic.com/s/article/####[0m

Database Specs:
```

```
 SL1 Architecture: Extensible

SL1 Specs:
 SL1 Release: EM7 12.2.0 [build 3100]
 ISO Release: 12.2.0

OS Specs:
 OS Release: Oracle Linux Server release 8.9
 MUD Deployment: No
 Hostname: ot1-monsldb01b
 Appliance Model: DB
 Active IPs:
  - 10.123.21.131
  - 10.123.29.149
 Licensed IP: 10.123.21.131
 Appliance Type: Virtual - VMware7,1
 Number of Cores: 4
 Total RAM: 24GB
 Disk Size: 299GB
 5 Minute Load Avg: 0.04
 Memory Utilization: 9%
 DRBD Version: 9.0.32-3
 DRBD Proxy: YES

DRBD Status:
r0 role:Secondary
  disk:UpToDate
  ot1-monsldb01a role:Primary
    peer-disk:UpToDate

[1;1;41m - DRBD SNMP agent extensions (/opt/em7/share/config/snmp_agent.d/50_db_compute.conf) - one or more of the stats files are missing - perhaps the snmpd service wasn't restarted?[0m

Other Information:

Checking for errors...
[1;1;41m - Nameserver 127.0.0.1 is not responding to DNS requests [KB 1222][0m
[1;1;41m - MySQL is NOT running[0m

Script results are also available in /tmp/ot1-monsldb01b_system_status_results.txt
All Done!
```

licensed state check

```
sudo licensed state
```

ha status check

```
sudo /opt/em7/backend/ha_status.py
```

Theory

file:/var/lib/em7/stats/sl1/licensed_state

TBD (see em-64413 and describe the role it played)

file:/var/lib/em7/stats/drbd_data/drbd_cs

TBD (see em-64413 and describe the role it played)

file:/opt/em7/share/config/coro_install.d/drbd_v9/drbd_proxy

TBD (See em-64409 and describe the role it played)

file:/opt/em7/share/config/coro_install.d/cib_stacked

TBD (See em-64907 and describe the role it played)

CRM Templates

See: Updating 10.2 CIB to remove Postfix resource and meta target-role=...

program:crm-template-updater.sh

See: SysEng: crm-template-updater

Post Mortem Articles

2025-06-13 EM-74609 RFO 00519148 - ongoing cpu issues after outage related to DR replication and DRBD network failures

Link: EM-74609: RFO 00519148 - ongoing cpu issues after outage related to DR replication and DRBD network failures CLOSED

From the description:

```
Observed Behavior:
When a failover occurs (DR not HA) CPU Utilization spikes up to ~%100. This appears to be related to DRBD

Expected Behavior:
While some CPU utilization increase would be expected post-failover, the amount of CPU utilization post-failover is worrysome to the customer.

Summary of troubleshooting steps taken so far and the current status:
I've taken a look at the logs for the last failover as well as performance data. I don't know of a reason that DRBD would need that much CPU.
I haven't not seen a bitmap or stream sync take up this many resources.
```

2024-10-30 EM-70932: DRBD constantly sync restarts at 0% after upgrade OL7 to OL8

ts_guide_to_jira_link

Link: EM-70932: DRBD constantly sync restarts at 0% after upgrade OL7 to OL8 CLOSED

From the description

```
Observed Behavior:
We have observed that when the Active DB is syncing with the DR node, however it is constantly resetting the sync process.
When checking the speeds, it would start quite high at dbdt1 500, where it would slowly creep down until the sync reset. After the sync reset, the speeds would hover around 0.05, a drastic decrease.
Summary of troubleshooting steps taken so far and the current status:
The was an issue when platform was on DRBD 8, and thus held off on resolving the issue and waited to upgrade to DRBD 9 which is is currently on.
system_status, drbd status, drbdproxy status, and cat /var/log/messages | grep drbd:

Active:
DR node:
Config active:
DR Node:

The current errors that we see on the Active DB are:
[272645.048339] drbd r0/0 drbd1: Congestion-fill threshold reached (653932 >= 573440)
drbd r0 gbip03-sl1db-01: conn( Connecting -> NetworkFailure )
drbd r0 gbip01-sl1db-01: Cluster is now split

Context:
gbip01-sl1db-01 = secondary HA
gbip02-sl1db-02 = DR
gbip03-sl1db-03 = active DB



We managed to help the issue by disconnecting the sync from the secondary HA to the DR, and then disconnected/reconnected the sync from the Active DB to the DR

 After this, the dbdt1 started at 1000, and stayed stable like that up until around 90% completed, and reset back to the behaviour seen before.

Every 1.0s: drbdsetup status --verbose --statistics        gbip02-sl1db-01: Tue Oct 15 16:05:07 2024
r0 node-id:2 role:Secondary suspended:no force-io-failures:no
  write-ordering:drain
 volume:0 minor:1 disk:Inconsistent backing_dev:/dev/mapper/em7vg-db quorum:yes
     size:903170292 read:384 written:1350624523 al-writes:95835
    bm-writes:2362263 upper-pending:0 lower-pending:1 al-suspended:no
    blocked:no
 gbip01-sl1db-01 node-id:1 connection:Connecting role:Unknown congested:no
    ap-in-flight:0 rs-in-flight:0
  volume:0 replication:Off peer-disk:DUnknown resync-suspended:dependency
     received:0 sent:0 out-of-sync:134025240 pending:0 unacked:0
 gbip03-sl1db-01 node-id:0 connection:Connected role:Primary congested:no
    ap-in-flight:0 rs-in-flight:0
  volume:0 replication:SyncTarget peer-disk:UpToDate done:49.52
     resync-suspended:no
```

```
    received:127143509 sent:0 out-of-sync:124115188 pending:88 unacked:0
    dbdt1:1026.96 eta:118

As well as this we changed the compression was proactively changed from lz4 level 10 to zlib level 5 to help with networking issues.
Also wanted to note the disk I/O as this has been tweaked before:
[root@gbip02-sl1db-01 ~]# cat /etc/my.cnf.d/silo_mysql.cnf | grep capacity
innodb_io_capacity = 3769
innodb_io_capacity_max = 5385

The config has been left alone when performing the above disconnects, as the congestion tuning is supposed to be optimized in DRBD 9, although when the issue appeared again we performed the following:
We tuned the memlimit to 1024M, and then set the congestion limit to 80% of that value. This still resulted in no change in behaviour.
Network, and Disk graphs:
```

2024-08-26 EM-66540: block drbd1: logical block size of local backend does not match (drbd:512, backend:4096)

ts_guide_to_jira_link

Link: EM-66540: block drbd1: logical block size of local backend does not match (drbd:512, backend:4096) CLOSED

From the description:

```
Summary:

Unable to sync. new Azure DR to existing HA cluster, Jason Clark is aware of the migration

Telstra is clustering Azure OL7 new DR node into existing OL7 on-prem HA cluster, getting into DRBD error when syncing indicating new DR disk block size of 4096 and existing HA node having 512. (logs attached)

95117+00:00 lxcnpaea3812db04 kernel: block drbd1: logical block size of local backend does not match (drbd:512, backend:4096); was this a late attach?
Line 43016: 2024-08-26T03:05:17.995287+00:00 lxcnpaea3812db04 kernel: block drbd1: logical block sizes do not match (me:512, peer:512); this may cause problems.
Line 43017: 2024-08-26T03:05:17.995377+00:00 lxcnpaea3812db04 kernel: block drbd1: drbd_sync_handshake:
Line 43018: 2024-08-26T03:05:17.995473+00:00 lxcnpaea3812db04 kernel: block drbd1: self 0334A93C77BD8BD0:0000000000000000:0000000000000000:0000000000000000 bits:24777905 flags:0
Line 43019: 2024-08-26T03:05:17.995559+00:00 lxcnpaea3812db04 kernel: block drbd1: peer 6792D7606A995AC1:0334A93C77BD8BD1:0333A93C77BD8BD1:0332A93C77BD8BD1 bits:24779512 flags:0
Line 43020: 2024-08-26T03:05:17.995646+00:00 lxcnpaea3812db04 kernel: block drbd1: uuid_compare()=-1 by rule 50
Line 43021: 2024-08-26T03:05:17.995727+00:00 lxcnpaea3812db04 kernel: block drbd1: Becoming sync target due to disk states.
```

2024-08-22 EM-66512: DR replication to DR node stops after HA failover

ts_guide_to_jira_link

Link: EM-66512: DR replication to DR node stops after HA failover CLOSED

From the description:

```
Observed Behavior: DRBD replication to DR node in HA+DR configuration doesn't restart after resources restart on HA secondary.

Expected Behavior: DRBD replication should always be connected between all 3 nodes and DR node should be UpToDate. Otherwise customer might not know that DR node contains obsolete data.

Summary of troubleshooting steps taken so far and the current status:
DRBD configuration is pretty much stock configuration.
All DB nodes were deployed using 12.2.1.1 ISO.
Deployed the stack to test steps on the disk resize. Wanted to make disk larger on VM, so shutdown DR and Secondary HA first, then HA Primary.
Resized the VM disk across all 3 DB nodes, and started Primary HA first, then Secondary HA, then DR, about 10-15 seconds one after the other. Prior to shutdown all DB nodes were in UpToDate state and all connected.
After a reboot HA secondary for some reason promoted itself, instead of HA primary.
Then I have noticed that DRBD reports HA nodes as UpToDate, but DR node is in "Connecting" and Inconsistent state.
No Split Brain messages, or issues found in /var/log/messages.
Tried to disconnect DRBD replication DR from HA02(current primary) using "drbdsetup disconnect r0 2" command, then reconnect back, but that did not resolve the issue.
Tried to use "drbdadm disconnect all" then "drbdadm connect all", that didn't help either.
Lab stack is HA01(10.64.70.80), HA02(10.64.70.88), DR(10.64.70.89).
Need help to identify the root cause and preventative measures.
Please see logs attached.
Thank you.
```

2024-08-20 EM-66492: em7 service gets hung after HA failover, doesn't restart for 15 minutes delaying other necessary services from starting

ts_guide_to_jira_link

Link: 💔EM-66492: em7 service gets hung after HA failover, doesn't restart for 15 minutes delaying other necessary services from starting DONE

From the description:

```
Observed Behavior: After hard stopping the HA primary, the em7 service appears to start on the other HA node, but then gets hung. About 15 minutes later it restarts and is able to start all the other services, like em7_event

Expected Behavior: It doesn't get hung the first time it starts, the other services start sooner than 15+ minutes later, so the product remains functional in a high availability failover

Summary of troubleshooting steps taken so far and the current status:
Tried to put the em7 process in debug to determine why it is getting hung, but the debug logs are not really showing any information.

Asked in a bunch of eng channels:
https://sciencelogic.slack.com/archives/C8XHFKRB7/p1724178435401989
https://sciencelogic.slack.com/archives/C1C2GTCAZ/p1724180664996849
```

```
https://sciencelogic.slack.com/archives/C04L444TLG2/p1724183550231199
```

2024-08-13 EM-66389: Sigterms and UI doesn't fully work after failover

ts_guide_to_jira_link

Link: EM-66389: Sigterms and UI doesn't fully work after failover CLOSED

From the description:

```
Observed Behavior: After upgrading to OL8, the customer had setup HA+DR cluster. When failing over, the node is not responding to DB IP or VIP for login. If tried with admin portal they are able to login without issue

Expected Behavior:  The VIP and DB IP should work for logging in.

Summary of troubleshooting steps taken so far and the current status:
We have collected the logs
crm services are up and running
collected the config files.
restarted the nginx,nextui,php-fpm services
```

2024-06-04 EM-65272: Setting up HA/DRBD

ts_guide_to_jira_link

Link: EM-65272: Setting up HA/DRBD CLOSED

From the description:

```
KBs:
13329
1259
Docs:
https://docs.sciencelogic.com/latest/Content/Web_Install_Configure/HA_DR_Configuration/disaster_recovery.htm
Confluence:
https://sciencelogic.atlassian.net/wiki/spaces/ENG/pages/28200075403/SL1+Corosync+Pacemaker+Clustering+v3.7+coro+install

Observed Behavior:
When attempting to use the ootb configuration scripts to setup the dr node, the script errors out without any appreciable error. Troubleshooting revealed that the drbd meta data was not written

Expected Behavior:
Should correctly configure dr node.

Summary of troubleshooting steps taken so far and the current status:
(Support Engineers need to provide a detailed summary of what steps have been taken so far and where they are currently at with the issue.)
The customer is currently deployed with a single node in OpenStack. They would like to leverage a DR Node in their new VMWare environment to migrate. The plan is to install the DR node in VMWare, sync DRBD to the VMWare node,
failover to the VMWare node, then decommision the Openstack node.

```

What we are facing is that initially, the customer was unaware that the DRBD filesystem needs to be the same size on both nodes. They had actually deployed the DR node much smaller. We added a drive, added it to the volume group,
extended the DRBD LV to be exactly the same number of extents as the primary node, and then used drbdadm to attempt to write the meta data to the drive. It was then that DRBD complained that the drives would need to be shrunk on
the openstack side to accomodate.

I then asked in #team-syseng and got a great response from Raynard and Jason about some changes to the DRBD Meta Data that I was unaware of. I then asked the customer to follow their recomendation of deploying the DR node
with ~10% more space than the primary.

Customer is reporting they have followed this recomendation and that the silo-cluster-install script is now erroring with the following error:
Starting DRBD proxy ...
Configuring DRBD proxy ...
Add connection ok.
trailing garbage 'level 10' ignored
WARN: nothing stacked for this host (sldb2-lab.stack71.netmd.ntw.bblabs), nothing to do in stacked mode!

[root@sldb2-lab log]# systemctl status drbd-proxy
● drbdproxy.service - LSB: Starts and stop the DRBD proxy.
Loaded: loaded (/etc/rc.d/init.d/drbdproxy; generated)
Active: failed (Result: exit-code) since Mon 2024-06-03 18:48:30 UTC; 5min ago

Docs: man:systemd-sysv-generator(8)
Process: 110900 ExecStop=/etc/rc.d/init.d/drbdproxy stop (code=exited, status=0/SUCCESS)
Process: 110911 ExecStart=/etc/rc.d/init.d/drbdproxy start (code=exited, status=1/FAILURE)
Jun 03 18:48:25 sldb2-lab.stack71.netmd.ntw.bblabs su[110920]: pam_unix(su-l:session): session opened for user drbdpxy>
Jun 03 18:48:25 sldb2-lab.stack71.netmd.ntw.bblabs su[110920]: pam_unix(su-l:session): session closed for user drbdpxy
Jun 03 18:48:25 sldb2-lab.stack71.netmd.ntw.bblabs drbdproxy[110911]: Configuring DRBD proxy ...
Jun 03 18:48:25 sldb2-lab.stack71.netmd.ntw.bblabs drbdproxy[110946]: Add connection ok.
Jun 03 18:48:30 sldb2-lab.stack71.netmd.ntw.bblabs drbdproxy[110945]: Command '/sbin/drbd-proxy-ctl -c set memlimit r0>
Jun 03 18:48:30 sldb2-lab.stack71.netmd.ntw.bblabs drbdproxy[110965]: WARN: nothing stacked for this host (sldb2-lab.s>
Jun 03 18:48:30 sldb2-lab.stack71.netmd.ntw.bblabs drbdproxy[110911]: Error configuring DRBD proxy, drbdutils package >
Jun 03 18:48:30 sldb2-lab.stack71.netmd.ntw.bblabs systemd[1]: drbdproxy.service: Control process exited, code=exited >
Jun 03 18:48:30 sldb2-lab.stack71.netmd.ntw.bblabs systemd[1]: drbdproxy.service: Failed with result 'exit-code'.
Jun 03 18:48:30 sldb2-lab.stack71.netmd.ntw.bblabs systemd[1]: Failed to start LSB: Starts and stop the DRBD proxy..
I've scheduled a meeting with the customer so I can at least validate that the customer followed the recomendation corectly and to check and see if the issue is the same. The error appears different nw as previusly it was c
omplaining that there was no resource defined for DRBD.
If engineering would like to be included on the meeting please let me know. Otherwise I will report back what I find on the meeting.

2024-05-09 EM-64907: CRM Template Outdated blocker for OL8 conversion

ts_guide_to_jira_link

Link: EM-64907: CRM Template Outdated blocker for OL8 conversion CLOSED

From the description:

```
Observed Behavior: CRM template out of date error not fixing
Expected Behavior: CRM template updater script failed to fix

Summary of troubleshooting steps taken so far and the current status:
Capgemini is in the process of upgrading 12.1.1 OL7 to OL8, Secondary DB has been Rebuilt. Primary DB is still on OL7
system status is showing errors about CRM template outdated. As per the docs CRM template should be updated to latest as a prerequisite.

Hence, Customer Followed the KBA steps and CRM template updater script, It failed to fix and error as below. The generated shadow file to commit was having incorrect information. If we commit that file it made the resources to go ORPHANED.
[root@in-mum-sldb02 crm_template_updater_script]# sudo bash crm-template-updater.sh --update
ERROR: configure.show-property: No such command
Putting cluster into maintenance...
[20240509 05:57:16]: Running: '_crm_live config property maintenance-mode=true'...
Creating a backup of the live CIB configuration...
[20240509 05:57:17]: Running: '_crm_live config save xml /root/cib_backups/live_cib_1715234232.xml'...
[20240509 05:57:17]: Running: '_crm_live config save /root/cib_backups/live_cib_1715234232.cli'...
Creating a shadow CIB to stage our changes to...
[20240509 05:57:18]: Running: '_crm_live cib new shadow_1715234232'...
INFO: cib.new: shadow_1715234232 shadow CIB created
[20240509 05:57:18]: Running: '_crm_shadow config load replace /opt/em7/share/config/coro_install.d/cib_stacked'...
INFO: modified colocation:colocation-drbd_r0-with-drbd_r0-L from ms_drbd_r0 to p_drbd_r0
INFO: modified colocation:colocation-g_em7-with-ms_drbd_r0 from ms_drbd_r0 to p_drbd_r0
INFO: modified order:order-drbd_r0-L-before-drbd_r0 from ms_drbd_r0 to p_drbd_r0
INFO: modified order:order-ms_drbd_r0-before-g_em7 from ms_drbd_r0 to p_drbd_r0
ERROR: could not open /opt/em7/share/config/coro_install.d/cib_stacked: [Errno 2] No such file or directory: '/opt/em7/share/config/coro_install.d/cib_stacked'
[20240509 05:57:19]: ERROR: Execution of '_crm_shadow config load replace /opt/em7/share/config/coro_install.d/cib_stacked' returned a non-zero (1) exit status.
[20240509 05:57:19]: Running: '_crm_shadow config property maintenance-mode=true'...
Setting: virtual_ip ip=10.246.101.45...

Error performing operation: No such device or address
Setting: virtual_ip cidr_netmask=26...

Error performing operation: No such device or address
Creating a diff of the shadow and live CIB configurations...
[20240509 05:57:20]: Running: '_crm_shadow cib diff'...
Saving a copy of shadow_1715234232 to /root/cib_backups/shadow_1715234232.cli...
[20240509 05:57:20]: Running: '_crm_shadow config save /root/cib_backups/shadow_1715234232.cli'...

Proceed with proposed changes?

Type 'Yes, update my cluster configuration.' to proceed: yes
```

```

The shadow CIB shadow_1715234232 has been saved with the diff/"proposed" changes in: /root/cib_backups/shadow_1715234232_diff_diff.txt
After reviewing the changes, to commit them to the running configuration run: 'crm -c shadow_1715234232 cib commit'
After committing, the cluster will be in maintenance. To bring the cluster out of maintenance run: 'crm config property maintenance-mode=false'
Then run  and run 'diff -bB <(sort /tmp/.crm_template) <(sort /tmp/.crm_running)' to see
if the changes performed resolved the issue(s).
It's safe to delete shadow_1715234232 afterwards; simply run: 'crm -c live cib del shadow_1715234232'
If this cluster was built before the 10.2 release (this system was ISO'ed with EM7 8.14.0-1587), the postfix resource would
probably be removed, and some additional step(s) might be required. Each node in the stack (including the DR node) would need the
following run: '/usr/bin/systemctl enable --now {postfix,em7_mailparse}'

Note:
HA clusters only need one node to be updated and the changes will be synchronized to other node. HA+DR stacks require
only one of the HA nodes and the DR node to be updated. DR stacks will need the update on the primary and DR nodes.

Exiting...
Captured the CRM configure show and Execution outputs for review.
```

2024-05-08 EM-64840 Issue after update CRM template

ts_guide_to_jira_link

Link: EM-64840: Issue after update CRM template CLOSED

From the description:

```
Observed Behavior: Issue after update CRM template
Expected Behavior: Issue after CRM template update cluster is not working
Stack: unknown
Current DC: eu1d115lpeaysldbp1 (version unknown) - partition with quorum
Last updated: Mon May 6 07:55:50 2024
Last change: Mon May 6 07:44:10 2024 by root via cibadmin on eu1d115lpeaysldbp1
2 nodes configured
0 resources configured
Online: [ eu1d115lpeaysldbc1 eu1d115lpeaysldbp1 ]

Active resources:
virtual_ip   (ocf::heartbeat:IPaddr2):    ORPHANED Started eu1d115lpeaysldbp1
p_drbd_r0    (ocf::linbit:drbd):   ORPHANED Master[ eu1d115lpeaysldbp1 eu1d115lpeaysldbc1 ]
mysql  (ocf::sciencelogic:mysql-systemd):    ORPHANED Started eu1d115lpeaysldbp1
p_fs_drbd1   (ocf::heartbeat:Filesystem):   ORPHANED Started eu1d115lpeaysldbp1
Summary of troubleshooting steps taken so far and the current status:
All resources are in ORPHANED state.
UI is up and running.
DR In sync.
```

2024-04-17 EM-64479: silo-cluster-install failes to create DRBD metadata on secondary and DR nodes,

ts_guide_to_jira_link

Link: EM-64479: silo-cluster-install failes to create DRBD metadata on secondary and DR nodes, CLOSED

From the description:

```
Observed Behavior: During the silo-cluster-install script the drbd metadata was not created on secondary and DR nodes. The script completed without error but the drbd connections would never connect

Expected Behavior: The script should determine there is an issue and either prompt what to fix or exit with an error so customers do not get into the state where
DRBD is running but nothing is connecting without any errors/notices/.
```

2024-04-15 EM-64415 /root/.cluster_install/r0.res:126: Parse error: 'ssocks | sdp | ipv4 | ipv6 | <ipv4 address> ' expected, but got ':'

ts_guide_to_jira_link

Link: EM-64415: /root/.cluster_install/r0.res:126: Parse error: 'ssocks | sdp | ipv4 | ipv6 | <ipv4 address> ' expected, but got ':' CLOSED

From the description:

```
Observed Behavior: Could not configure DR config after OL8 migration
Expected Behavior: Ability to configure DR config after OL8 migration
Summary of troubleshooting steps taken so far and the current status:
Jason had just found this same bug this morning. He provided the workaround here: https://sciencelogic.slack.com/archives/C8XHFKRB7/p1713204078584909?thread_ts=1713203550.732069&cid=C8XHFKRB7
That workaround was applied to this customer stack. I am not sure if a bug type jira was created or not yet, but I would like to write a known issue KB article documenting this defect so that
if other customers hit the issue, the customer or support can find the KB and the workaround by just searching for the error message.
```

2024-04-15 EM-64413: SNMP reporting that one DB is not licensed and another is not in sync, neither of those appear to be true

ts_guide_to_jira_link

Link: EM-64413: SNMP reporting that one DB is not licensed and another is not in sync, neither of those appear to be true CLOSED

From the description:

```
Observed Behavior: The /var/lib/em7/stats/sl1/licensed_state file on the HA passive and the /var/lib/em7/stats/drbd_data/drbd_cs file on the DR node are not being updated correctly

Expected Behavior: Those files contain the correct state information
```

```

Summary of troubleshooting steps taken so far and the current status:

Issue the customer is reporting is they are getting events for one appliance not being licensed, and another node not being in sync. Tracing the issues back, the EM7 SNMP OID is not providing the correct
information. It appears that whatever process is supposed to update that file is not updating that file for some reason. Checked the owner and permissions of the files on both nodes, they appear correct.
Need assistance determining why these files are not updating.

https://sciencelogic.slack.com/archives/C1FGDQN2C/p1713192017830859 is the first #support thread I made while trying to figure out the above. After we got that figured out, I created a slack channel for
this issue here: https://sciencelogic.enterprise.slack.com/archives/C06V18G57SL

That second slack link is a channel that is very fresh and contains only the most relevant of information. I will provide two screenshots from that channel on the escalation after I create it - one from
each device showing that the config file does not match the output of other commands(like drbdadm status, and system_status).
```

2024-04-15 EM-64409: coro_install.d/drbd_v9/drbd_proxy template is incorrect

ts_guide_to_jira_link

Link: 🐞 EM-64409: coro_install.d/drbd_v9/drbd_proxy template is incorrect CLOSED

From the description:

```
/opt/em7/share/config/coro_install.d/drbd_v9/drbd_proxy file has incorrect values resulting in bad substitutions.  This prevents DR systems from working.
```

2024-04-04 EM-64174: Silo-cluster-install script fails to connect to the remote database when the password contain special chars

ts_guide_to_jira_link

Link: 💔 EM-64174: Silo-cluster-install script fails to connect to the remote database when the password contain special chars CLOSED

From the description:

```
Observed Behavior: When running the cluster install script the script gets unable to connect to mysql with user root when the password contains special chars even though using silo_mysql -h from the command line works using the same password.

Expected Behavior: The script silo-cluster-install script should support passwords with special chars

Summary of troubleshooting steps taken so far and the current status:

I tried using single/double quotes around the password when entering it into the script prompt
I updated the silo.conf with the root user and correct password using visilo but the script still does not pass the password correctly.
```

```

Workaround was supplied by Jason

Make a copy of the file /opt/em7/share/scripts/environment.d/cluster_install_helper_functions.sh
cp /opt/em7/share/scripts/environment.d/cluster_install_helper_functions.sh /opt/em7/share/scripts/environment.d/cluster_install_helper_functions.old
Edit the file /opt/em7/share/scripts/environment.d/cluster_install_helper_functions.sh using vi and change the line below
 printf -v _l_pass "%q" "${_SILO_DBPASSWD}" to  printf -v _l_pass "%s" "${_SILO_DBPASSWD}" - so your changing %q to %s
save the file
Rerun the silo-cluster-install script.
```

2024-04-04 EM-64157: DRBD replication does not work after phase 2 due to "The peer's disk size is too small!"

ts_guide_to_jira_link

Link: EM-64157: DRBD replication does not work after phase 2 due to "The peer's disk size is too small!" CLOSED

From the description:

```
Observed Behavior: During migration to OL8, customer used HA secondary as OL8 target node. Phase 2 completed fine, but DRBD replication was not connecting during phase 3. Customer opened a case to
investigate and we found out that HA Primary(OL7 source) had "The peer's disk size is too small!" even though logical volumes on both source and destination had the same number of extents. We had to
extend logical volume on OL8 target and re-create metadata, at which point DRBD was able to replicate from OL7 source.

Expected Behavior: We shouldn't have "The peer's disk size is too small!" issue during migration.
```

2024-03-27 EM-64054: DRBD is triggered and becomes out-of-sync. This process gets hung and doesn't complete.

ts_guide_to_jira_link

Link: EM-64054: DRBD is triggered and becomes out-of-sync. This process gets hung and doesn't complete. CLOSED

From the description:

```
Observed Behavior: The customer is running 12.1.1 with OL8 which had DRBD version: 9.0.32. They are seeing an issue where the DR node never fully sync's. They can resolve the issue by restarting
drbdproxy, however that is not a good long-term solution and does not always work as expected.

Expected Behavior: There should not be any synchronization issues between the Primary and DR other than network and/system. They are seeing resyncs and failures to sync.
```

From this comment EM-64054: DRBD is triggered and becomes out-of-sync. This process gets hung and doesn't complete. CLOSED

```
I’ve gone through all of the data supplied and there appear to be a number of problems with this system.  I strongly suspect that all of these problems are are contributing to the symptoms that are being seen.

This system appears to have been initially deployed at a much small size than what is currently running.  Because configurations are generated dynamically, the recommendation is to deploy a CDB as the size it’s expected to run
at and not take a “scale up” approach.  When scaling up a CDB there’s number of things that must be taken in to account to do it correctly.

Here are the things that need to be reviewed and fixed:

    innodb_buffer_pool_size
    This is currently set to 8GB in the pt-stalk i was given, yet this system has 132GB of RAM in it.  This is far too low of a size.  It suggests the system was deployed at 10GB or 12GB of ram initially.  Having this value too
    small places excessive IO load on the disk and severely impacts SL1 performance.
    Using vimysql, delete this line from the configuration and save the file.  It should regenerate the line at the default recommendation.

    innodb_log_file_size
    This is currently set to 1GB in the pt-stalk I was given. This is much too small for a system of this size.  Having this value too small causes heavy impact to disk IO as the system will continually force flush data to the disk
    instead of properly consolidating IO’s
    I believe there is a KB that has instructions on how to change this.  It is not a simple as just editing the silo_mysql.cnf file.  Please follow those instructions and size it to 10GB.

    innodb_io_capacity and innodb_io_capacity_max
    These are currently at their default values.  The defaults are too low for an SL1 system to operate effectively.
    Using vimysql set the values to 3000 and 5000 respectively.  These will be better values that are likely in the performance range of their disk.

    memlimit in DRBD proxy configuration
    This is currently set to 400MB which is the default for a very small system.  A system with 132GB of RAM should have much higher value.  With such a low memory value the proxy is going to fill up and congest on its memory which w
    ill have unintended side effects (such as making the proxy misbehave).
    Set this value to 2048M, this would be our default for systems that have more than 128GB of RAM.  This is a tunable value based upon load but this should be a good value for this system.  We can reevaluate if issues persist.

    congestion-fill in DRBD configuration
    This is currently set to 280M which is the default for a very small system.  This will cause DRBD to become congested too quickly and forced disconnections.
    Set this value to 1434M, this would be our default for systems that have more than 128GB of RAM.  This is a tunable value based upon load but this should be a good value for this system.  We can reevaluate if issues persist.

    The system is not using vNUMA.  I don’t have any direct knowledge of the underlying ESXi system, so this may not apply, but in a general sense a system of 16 cores, and 132GB of RAM is highly likely to need vNUMA enabled so the
```

```
    guest can optimally use the underlying hardware.  This needs to be reviewed by their VMware administrators to determine if action is needed.

I believe addressing these 6 issues will put the system in to a much better state and very likely to resolve the issues seen with DRBD.
```

2024-02-09 EM-63276 Customer Needs Advice on Resizing DB disk space beyond 2TB with special circumstances

ts_guide_to_jira_link

Link: EM-63276: Customer Needs Advice on Resizing DB disk space beyond 2TB with special circumstances CLOSED

From the description:

```
Hi team,

Customer provided background and the steps in their plan so I'm just pasting it below. Kindly advise on this plan and if we need to alter their course of action in any way.

we are looking for advice on resizing our DB disk space. The DB will grow to target size of 3 TB, but currently it's limited by DOS partitioning scheme, where the partition maximum size is 2 TB.
Furthermore, our DB is in HA setup. I found your KB article 9367 which mentions that appliance reinstall is preferred to converting disk to GPT. We are fine to do that, but would like to check
with you our plan, particularly concerning DRBD/HA sync.

The plan is following:

1. start with DB active on the primary node, put the cluster in maintenance
2. reinstall secondary node with appliance having 3TB disk (according to mentioned KB article this should automatically create GPT partitioning scheme)
3. configure HA (standby role) on the secondary node
4. stop the cluster maintenance and sync the DB to the standby node (meaning that the DB will be synced to a larger filesystem, but will this work with unequally sized filesystems?)
5. make the DB active on the standby node, put the cluster in maintenance
6. reinstall primary node with appliance having identical 3TB disk
7. configure HA (primary role) on the primary node
8. change the role of the node to standby (so it can be synced back from currently active secondary node ... I suppose this is needed, but how exactly we should do that?
9. finally, remove the cluster maintenance
10. in the end, both nodes will have equally sized disks and DB filesystems.
```

2024-01-31 EM-63149: Coro_config not working after 12.1.1. update

ts_guide_to_jira_link

Link: EM-63149: Coro_config not working after 12.1.1 update CLOSED

From the description:

```
After rebooting the primary node in a DR topology after updating to 12.1.1, customer found they couldn't run coro_config to promote Corosync to
```

```
restore uptime. The error was occurring on both nodes, Primary and DR. This is a similar case to
```

💔 EM-62888: coro_config fails with error CLOSED

2020-04-28 Symantech

Article here: Symantech DRBD HA.